



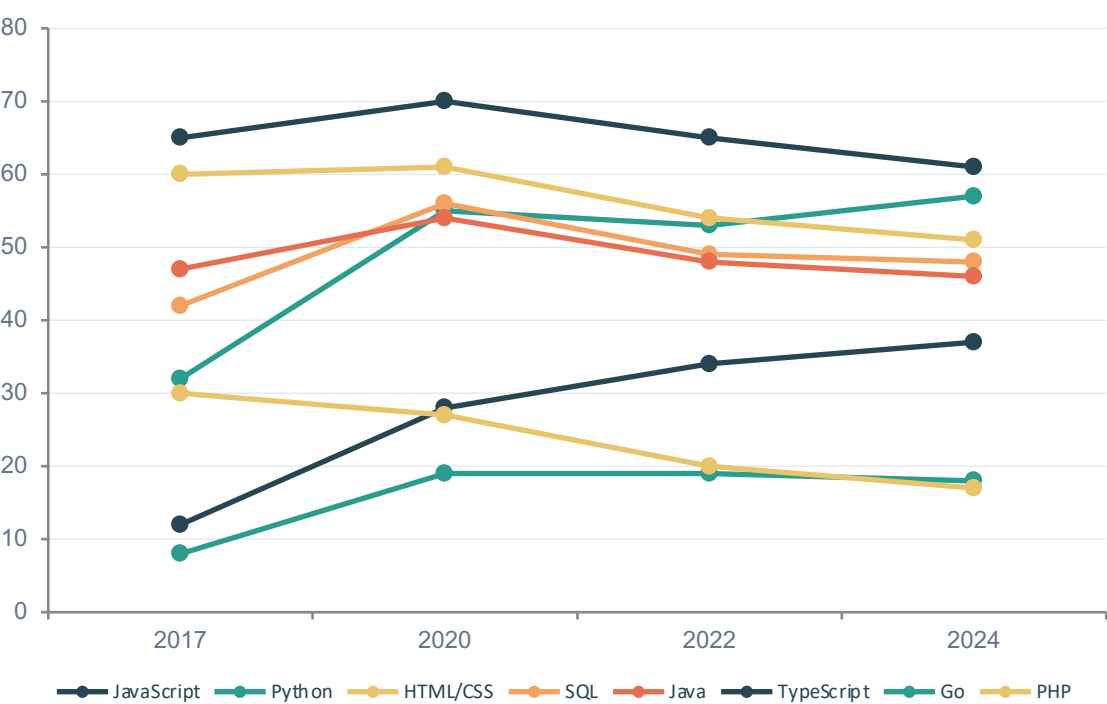
State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

Based on JetBrains' annual survey of 23,262 developers worldwide

2024

Python Closes the Gap: 8 Years of Language Usage Shifts

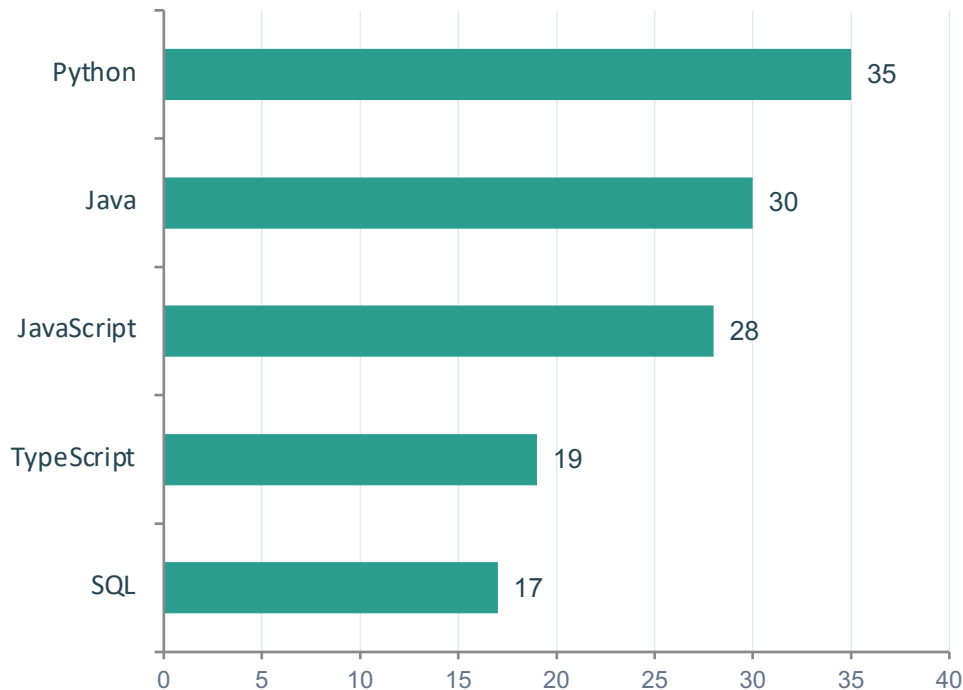


Language	2024	Trend
JavaScript	61%	Declining from peak
Python	57%	Steady growth
HTML/CSS	51%	Gradual decline
SQL	48%	Plateaued
Java	46%	Slow decline
TypeScript	37%	Strong growth
Go	18%	Plateaued
PHP	17%	Significant decline
Kotlin	14%	Plateaued
Rust	11%	Steady growth

Source: JetBrains State of Developer Ecosystem 2024

Python rose from 32% to 57%; TypeScript tripled from 12% to 37%

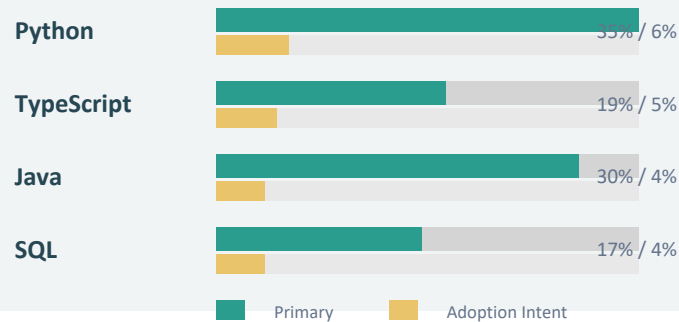
Python Is Now the #1 Primary Language at 35%



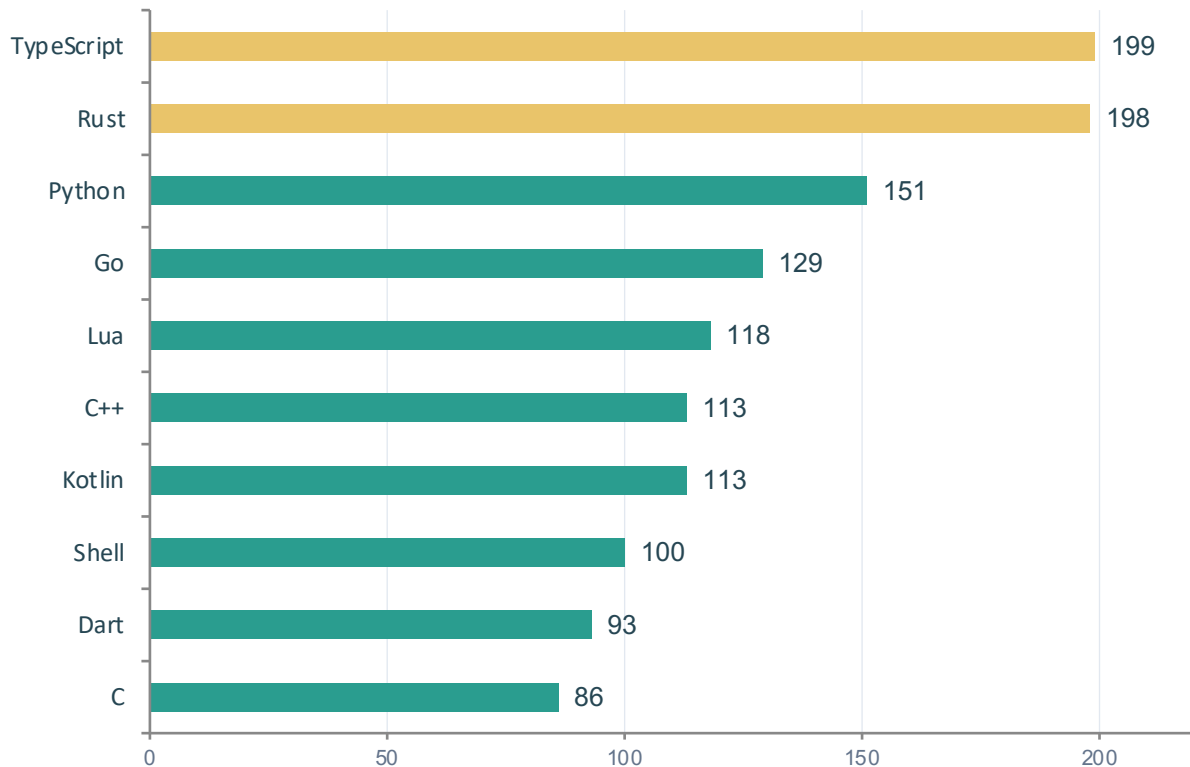
Ecosystem Lock-In

Adoption intent remains below 6% across all languages — developers rarely plan to switch their primary stack.

Adoption Intent vs. Primary Usage



TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Top 2: Nearly Tied
TypeScript 199
Rust 198

Difference: 1 point

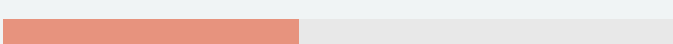
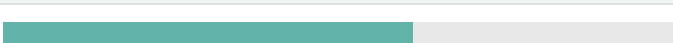
Why They Lead

Type-safe, modern-tooling languages dominate the top spots.

Both TypeScript and Rust combine:

- Strong type systems
- Modern developer tooling
- Expressive syntax

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Leading Language	Share	Dominance
AI / Machine Learning	Python	31%	
Data Processing & Analytics	Python	17%	
User Interface	TypeScript	62%	
APIs & Services	Java	49%	
Application Logic & Workflows	Java	55%	
System Tools & Components	C++	27%	
Integrating with APIs & Services	Python	38%	

Language choice is driven by domain, not general preference — hiring and training should align with domain specialization.

What This Means for Your Technology Strategy in 2025

- 1 Invest in Python for AI/ML teams — it is the undisputed leader
- 2 TypeScript is the future of frontend and full-stack
- 3 Rust and Go are the growth plays for infrastructure and systems
- 4 PHP and Ruby hiring pools are shrinking — plan succession now
- 5 Developer satisfaction correlates with type safety and modern tooling — factor this into retention strategy